

Project 3

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(a)

Prices become simple monthly returns, $R_t = P_t/P_{t-1} - 1$. 216 prices give 215 returns for each of the 5 stocks.

```
#> first month: 1986-01    last month: 2003-12
```

```
#> returns: 215 months x 5 stocks
```

(b)

Table 1: Mean monthly return and standard deviation, 1986-02 to 2003-12

Stock	Ticker	Mean (%)	Std. dev. (%)
Exxon-Mobil	XOM	0.2763	7.6178
General Motors	GM	0.3583	9.7256
Hewlett Packard	HPQ	0.6623	12.7646
McDonalds	MCD	0.0454	9.7955
Boeing	BA	0.4568	9.6138

Table 2: Variance-covariance matrix

	XOM	GM	HPQ	MCD	BA
XOM	0.005803	0.001389	0.001667	0.000790	0.001351
GM	0.001389	0.009459	0.003945	0.002281	0.002579
HPQ	0.001667	0.003945	0.016294	0.002864	0.001470
MCD	0.000790	0.002281	0.002864	0.009595	0.003211
BA	0.001351	0.002579	0.001470	0.003211	0.009242

(c)

With only two stocks the minimum risk portfolio follows from setting $d\sigma_p^2/dx = 0$, where x is the weight on Exxon-Mobil:

$$x_{XOM} = \frac{\sigma_{BA}^2 - \sigma_{XOM,BA}}{\sigma_{XOM}^2 + \sigma_{BA}^2 - 2\sigma_{XOM,BA}}, \quad x_{BA} = 1 - x_{XOM}$$

```
#> x_XOM = 0.639315
```

```
#> x_BA = 0.360685
```

```
#> E = 0.003414 (0.3414% per month)
```

```
#> sd = 0.064787 (6.4787% per month)
```

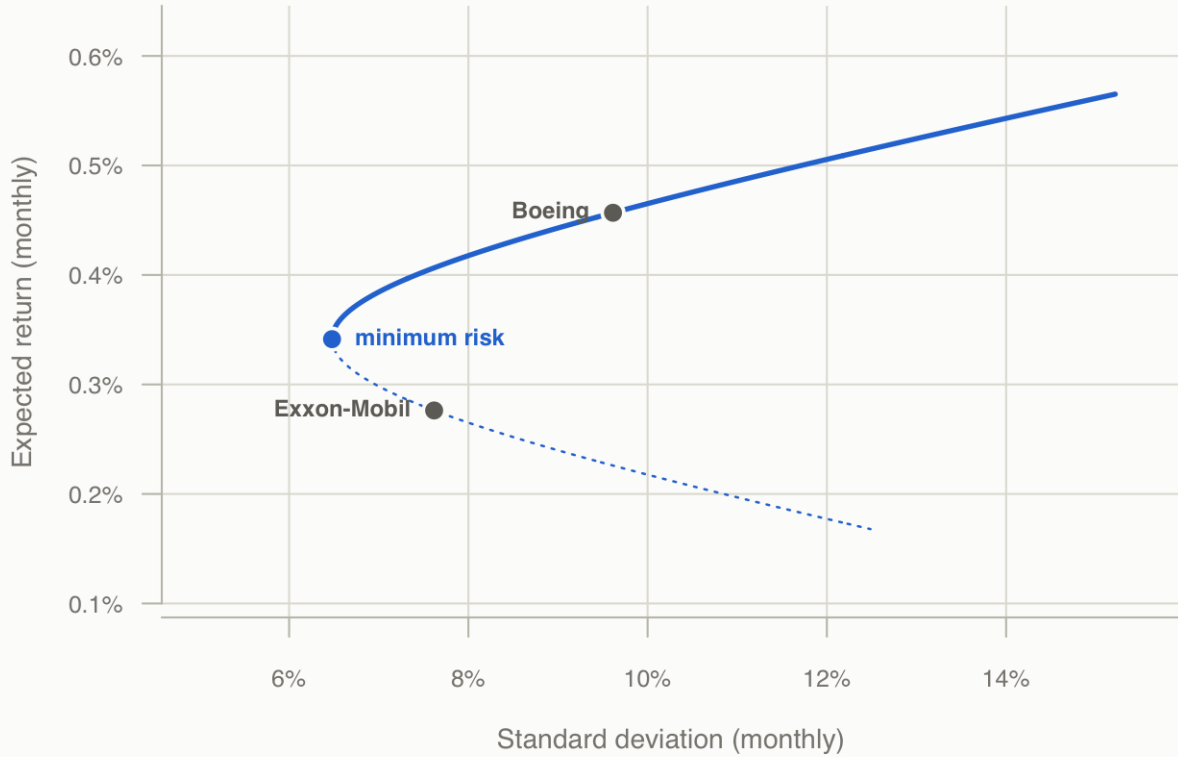
Both weights are positive, so no short selling is needed to reach it. The resulting standard deviation, 6.48%, is below either stock on its own (Exxon-Mobil 7.62%, Boeing 9.61%); the two are only weakly correlated ($\rho = 0.184$), so combining them removes risk.

(d)

Sweeping x_{XOM} from -0.6 to 1.6 traces the portfolio possibilities curve. The efficient frontier is the part of the curve at or above the minimum risk portfolio: for any point on the dotted lower arm there is a point directly above it with the same standard deviation and a higher expected return, so no one would hold it.

(c)-(d) Portfolio possibilities curve, Exxon-Mobil and Boeing

Solid arm is the efficient frontier; dotted arm is dominated by it.



(e)

The 2499 supplied combinations of x_a, x_b, x_c with $x_a + x_b + x_c = 1$ give the cloud plotted in the next section.

```
#> portfolios: 2499
```

```
#> E ranges -0.260% to 1.543% per month
```

```
#> sd ranges 5.974% to 41.977% per month
```

Many of the supplied weights lie well outside $[0, 1]$ — they run from -2.08 to 4.60 — which is what short sales allow, and why the cloud reaches standard deviations far above any individual stock.

(f)

With a risk free asset the investor holds some mix of R_f and one risky portfolio, so the best risky portfolio is the one maximising the slope $(\bar{R}_p - R_f)/\sigma_p$. Maximising that ratio gives

$$\mathbf{z} = \Sigma^{-1}(\bar{R} - R_f \mathbf{1}), \quad \mathbf{x}_G = \frac{\mathbf{z}}{\sum_i z_i}$$

the rescaling being what forces the weights to sum to one.

Table 3: Composition of the point of tangency G , $R_f = 0.001$

Stock	Weight
Exxon-Mobil	0.528478
McDonalds	-0.495588
Boeing	0.967110

```
#> E      = 0.005652  (0.5652% per month)
#> sd      = 0.102526  (10.2526% per month)
#> slope = (E - Rf)/sd = 0.045378
```

(g)

The risk free asset has no variance and no covariance with anything, so a mix of w in G and $1 - w$ in R_f gives

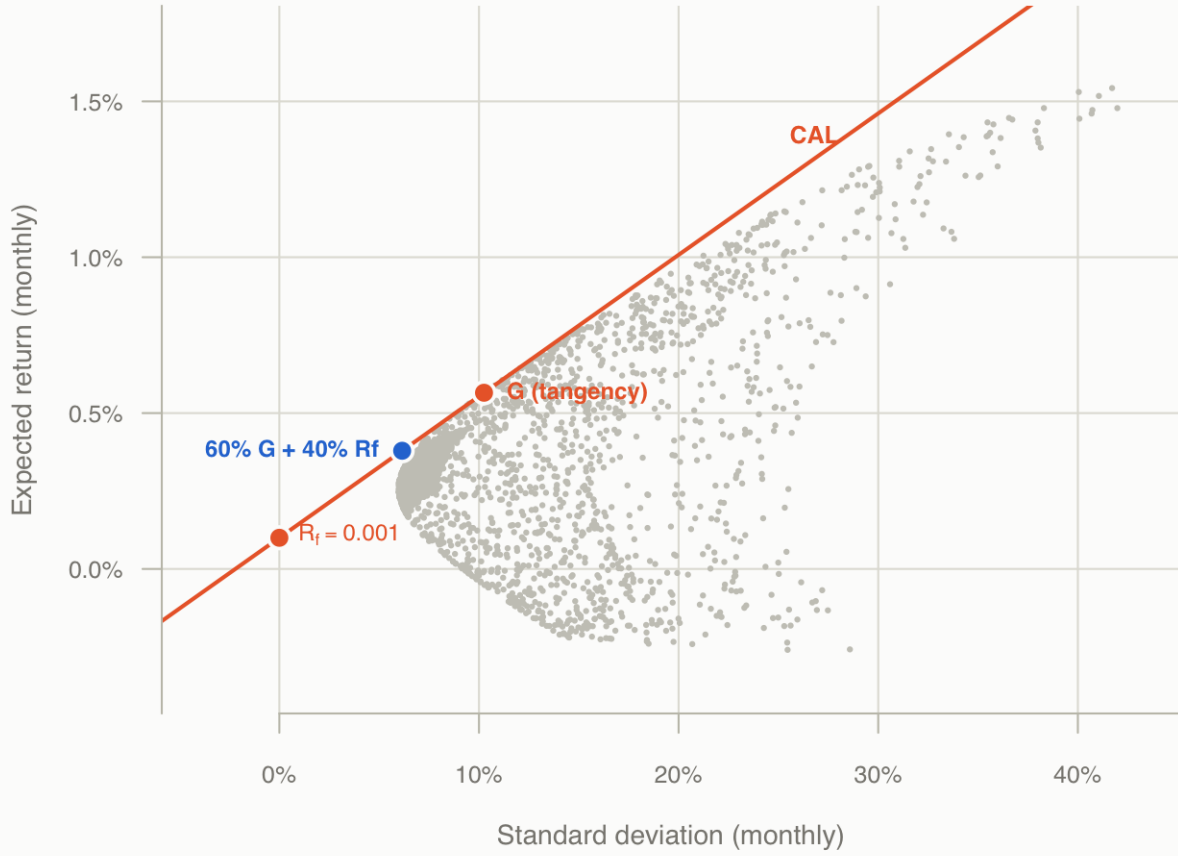
$$\bar{R}_p = w\bar{R}_G + (1 - w)R_f, \quad \sigma_p = w\sigma_G$$

```
#> E  = 0.6(0.005652) + 0.4(0.0010) = 0.003791  (0.3791% per month)
#> sd = 0.6(0.102526)                = 0.061515  (6.1515% per month)
```

Because $w < 1$ the position is a lending one — 40% of the money sits in the risk free asset — so it falls on the CAL between R_f and G , marked in blue below.

(e)-(g) Cloud, point of tangency, and the capital allocation line

2499 combinations of Exxon-Mobil, McDonalds and Boeing with weights summing to 1.



(h)

Table 4: $\mathbf{x}$ evaluated at $E = 0.003791$, the expected return from (g)

Stock	$\mathbf{x}$	$0.6 * G$
Exxon-Mobil	0.317087	0.317087
McDonalds	-0.297353	-0.297353
Boeing	0.580266	0.580266

```
#> sum(x)      = 0.600000
```

```
#> 1 - sum(x) = 0.400000
```

x represents the allocation to each of the three stocks in the complete portfolio from (g). They sum to 0.6, and the missing 0.4 is the fraction held in the risk free asset.

Substituting $E = w\bar{R}_G + (1 - w)R_f$ into

$$\mathbf{x} = \frac{(E - R_f) \Sigma^{-1}(\bar{\mathbf{R}} - R_f \mathbf{1})}{(\bar{\mathbf{R}} - R_f \mathbf{1})' \Sigma^{-1}(\bar{\mathbf{R}} - R_f \mathbf{1})}$$

gives $\mathbf{x} = w \mathbf{x}_G$ algebraically.

(i)

Short sales are still allowed, but the risk free asset is gone.

1

Running the tangency formula at two different values of R_f gives two portfolios that both sit on the frontier. The one for $R_{f1} = 0.001$ is G from part (f).

Table 5: A ($R_f = 0.001$) and B ($R_f = 0.002$)

Stock	A	B
Exxon-Mobil	0.528478	0.531220
McDonalds	-0.495588	-1.802663
Boeing	0.967110	2.271443

```
#> A: E = 0.005652 (0.5652%)    sd = 0.102526 (10.2526%)
```

```
#> B: E = 0.011024 (1.1024%)    sd = 0.236554 (23.6554%)
```

Raising the assumed risk free rate pushes the tangency point up and to the right along the frontier: B is a far more aggressive portfolio, shorting McDonalds 1.80 to hold 2.27 of Boeing.

2

$$\sigma_{AB} = \mathbf{x}'_A \Sigma \mathbf{x}_B$$

```
#> xA' Sigma xB = 0.0226482317
```

```
#> (C/D)(E_A - A/C)(E_B - A/C) + 1/C = 0.0226482317
```

```
#> correlation = 0.933838
```

The second line is the closed form for the covariance of two frontier portfolios (handout #11), computed from the constants A, B, C, D of Project 2. The two agree exactly, which confirms both A and B really are on the frontier rather than merely near it.

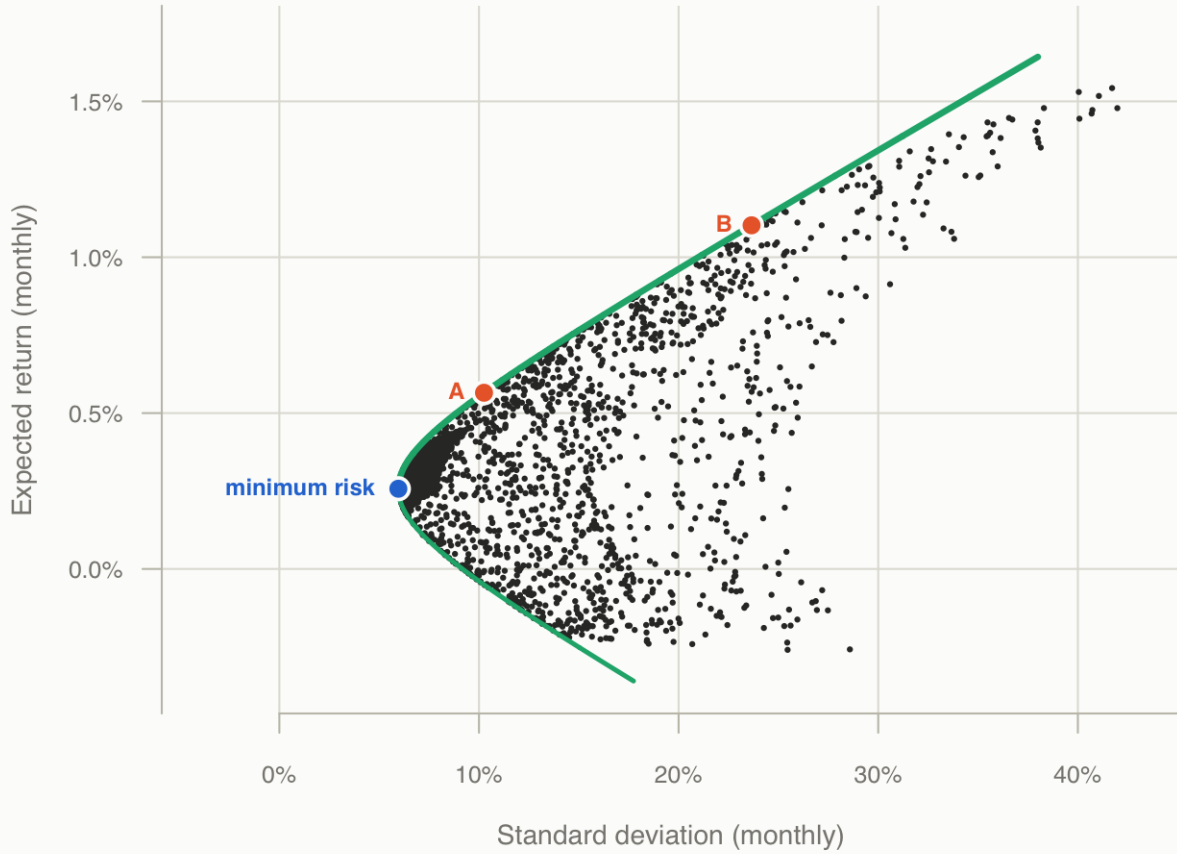
3

Two frontier portfolios span the whole frontier, so combining A and B — using no individual stock at all — traces it out:

$$\bar{R}_p = a\bar{R}_A + (1-a)\bar{R}_B, \quad \sigma_p^2 = a^2\sigma_A^2 + (1-a)^2\sigma_B^2 + 2a(1-a)\sigma_{AB}$$

(i) Efficient frontier traced from portfolios A and B

Green curve is combinations of A and B only; it envelopes the cloud from the left.



The green curve is generated purely from A and B , and it wraps the cloud from the left. Two checks:

```
#> max |two-fund mix - closed form frontier| over 2000 mixes: 2.50e-16
```

```
#> smallest gap between a cloud point and the frontier: 5.81e-08
```

The first confirms every mix of A and B lands on $\sigma^2 = (CE^2 - 2AE + B)/D$; the second confirms no point in the cloud gets inside the curve. The upper arm, above the minimum risk portfolio, is the efficient frontier and is drawn thicker.

4

$$\mathbf{x}_{min} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

Table 6: Minimum risk portfolio of the three stocks

Stock	Weight
Exxon-Mobil	0.526906
McDonalds	0.253653
Boeing	0.219440

```
#> E = A/C = 0.002573 (0.2573% per month)
```

```
#> sd = sqrt(1/C) = 0.059619 (5.9619% per month)
```

All three weights are positive here, unlike A and B : minimising variance alone ignores the mean returns, so there is no reason to short the low-return stock. Its standard deviation, 5.962%, is the leftmost point of the green curve and is smaller than any of the 2499 portfolios in the cloud.

Appendix: full code

```
local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "tangency.R")
    setwd(dirname(normalizePath(f)))
})

# ---- (a) prices to returns -----

a <- read.table("statc183c283_5stocks.txt", header = TRUE)
stopifnot(nrow(a) == 216, names(a)[1] == "date")

a <- a[order(a$date), ] # oldest -> newest
dates <- as.Date(as.character(a$date), format = "%Y%m%d")
P <- as.matrix(a[, -1])
colnames(P) <- c("XOM", "GM", "HPQ", "MCD", "BA")
stopifnot(!anyNA(P), !is.unsorted(dates))

n <- nrow(P)
ret <- P[-1, ] / P[-n, ] - 1 # 215 x 5 simple monthly returns
stopifnot(nrow(ret) == 215, ncol(ret) == 5)

# ---- (b) mean returns and variance-covariance matrix -----

Rbar <- colMeans(ret)
S <- cov(ret)
stopifnot(qr(S)$rank == 5)

cat("(b) mean monthly return, in percent:\n")
print(round(Rbar * 100, 4))
cat("\n variance-covariance matrix:\n")
print(round(S, 6))
cat("\n")

# ---- (c) minimum risk portfolio, Exxon-Mobil and Boeing -----
# Two assets, so the weight follows from setting  $d(\sigma^2)/dx = 0$ :
#  $x_{XOM} = (sd_{BA}^2 - cov) / (sd_{XOM}^2 + sd_{BA}^2 - 2 cov)$ 

stk2 <- c("XOM", "BA")
S2 <- S[stk2, stk2]
Rbar2 <- Rbar[stk2]

vx <- S2["XOM", "XOM"]; vb <- S2["BA", "BA"]; cxb <- S2["XOM", "BA"]
x2min <- c(XOM = (vb - cxb) / (vx + vb - 2 * cxb))
x2min["BA"] <- 1 - x2min["XOM"]
```



```

# The general form  $\Sigma^{-1} 1 / (1' \Sigma^{-1} 1)$  must give the same answer
stopifnot(all.equal(as.numeric(x2min),
                    as.numeric(solve(S2) %*% rep(1, 2) / sum(solve(S2))))))

E2min <- as.numeric(t(x2min) %*% Rbar2)
sd2min <- sqrt(as.numeric(t(x2min) %*% S2 %*% x2min))

# It has to be at least as good as holding either stock alone
stopifnot(sd2min < sqrt(vx), sd2min < sqrt(vb))

cat("(c) minimum risk portfolio, Exxon-Mobil and Boeing\n")
cat(sprintf("    x_XOM = %.6f    x_BA = %.6f\n", x2min["XOM"], x2min["BA"]))
cat(sprintf("    E = %.6f (%.4f%%)    sd = %.6f (%.4f%%)\n\n",
            E2min, E2min * 100, sd2min, sd2min * 100))

# ---- (d) portfolio possibilities curve for the two stocks -----

x2grid <- seq(-0.6, 1.6, length.out = 900)
W2      <- cbind(x2grid, 1 - x2grid)
E2c     <- as.numeric(W2 %*% Rbar2)
sd2c    <- sqrt(rowSums((W2 %*% S2) * W2))
stopifnot(min(sd2c) >= sd2min - 1e-12)    # nothing beats the minimum

# ---- (e) cloud of points, Exxon-Mobil / McDonalds / Boeing -----
# 2499 combinations of (xa, xb, xc) summing to 1, short sales allowed

abc <- read.table("statc183c283_abc.txt", header = TRUE)
# weights are stored to 6 decimals, so they sum to 1 only to ~1e-9
stopifnot(ncol(abc) == 3, max(abs(rowSums(abc) - 1)) < 1e-8)

stk3 <- c("XOM", "MCD", "BA")
Rbar3 <- Rbar[stk3]
S3    <- S[stk3, stk3]
S3inv <- solve(S3)
one3  <- rep(1, 3)

X3      <- as.matrix(abc)
E_cloud <- as.numeric(X3 %*% Rbar3)
sd_cloud <- sqrt(rowSums((X3 %*% S3) * X3))

cat("(e) cloud of", nrow(X3), "portfolios\n")
cat(sprintf("    E ranges %.5f to %.5f\n", min(E_cloud), max(E_cloud)))
cat(sprintf("    sd ranges %.5f to %.5f\n\n", min(sd_cloud), max(sd_cloud)))

# ---- (f) point of tangency G,  $R_f = 0.001$  -----
# Maximising the slope  $(E_p - R_f)/\sigma_p$  gives  $z = \Sigma^{-1} (Rbar - R_f 1)$ ,
# then rescale to weights:  $x = z / \text{sum}(z)$ 

Rf1 <- 0.001

tangency <- function(Rf) {
  z <- S3inv %*% (Rbar3 - Rf * one3)
  x <- as.numeric(z / sum(z))

```

```

    setNames(x, stk3)
}

xG      <- tangency(Rf1)
E_G     <- as.numeric(t(xG) %>% Rbar3)
sd_G    <- sqrt(as.numeric(t(xG) %>% S3 %>% xG))
slope_G <- (E_G - Rf1) / sd_G

stopifnot(abs(sum(xG) - 1) < 1e-12)
# G must have the highest Sharpe ratio in the whole cloud
stopifnot(slope_G >= max((E_cloud - Rf1) / sd_cloud))

cat("(f) point of tangency G, Rf =", Rf1, "\n")
print(round(xG, 6))
cat(sprintf("    E = %.6f (%.4f%%)    sd = %.6f (%.4f%%)    slope = %.6f\n\n",
            E_G, E_G * 100, sd_G, sd_G * 100, slope_G))

# ---- (g) 60% G, 40% risk free asset -----

wG      <- 0.6
E_g     <- wG * E_G + (1 - wG) * Rf1
sd_g    <- wG * sd_G                      # the risk free asset adds no variance

stopifnot(abs(E_g - (Rf1 + slope_G * sd_g)) < 1e-15) # sits on the CAL

cat("(g) 60% G + 40% risk free\n")
cat(sprintf("    E = %.6f (%.4f%%)    sd = %.6f (%.4f%%)\n\n",
            E_g, E_g * 100, sd_g, sd_g * 100))

# ---- (h)  $x = (E - R_f) \Sigma^{-1} (R_{bar} - R_f \mathbf{1}) / (R_{bar} - R_f \mathbf{1})' \Sigma^{-1} (\dots) -$ 

excess <- Rbar3 - Rf1 * one3
Hq      <- as.numeric(t(excess) %>% S3inv %>% excess)
x_h     <- as.numeric((E_g - Rf1) * S3inv %>% excess / Hq)
names(x_h) <- stk3

# x is the holding of each *stock* in the combined portfolio of (g): it is
# exactly 0.6 (the weight on G) times G's own composition, and what is left
# over, 1 - sum(x) = 0.4, is the risk free asset.
stopifnot(all.equal(x_h, wG * xG))
stopifnot(abs(sum(x_h) - wG) < 1e-12)
stopifnot(abs(as.numeric(t(x_h) %>% Rbar3) + (1 - wG) * Rf1 - E_g) < 1e-15)
stopifnot(abs(sqrt(as.numeric(t(x_h) %>% S3 %>% x_h)) - sd_g) < 1e-15)

cat("(h) x for E =", format(E_g, digits = 6), "\n")
print(round(x_h, 6))
cat(sprintf("    sum(x) = %.6f, so %.0f%% in the three stocks and %.0f%% risk free\n",
            sum(x_h), sum(x_h) * 100, (1 - sum(x_h)) * 100))
cat(sprintf("    max |x - 0.6 * xG| = %.2e\n\n", max(abs(x_h - wG * xG))))

# ---- (i) short sales allowed, no risk free asset -----

Rf2 <- 0.002

```

```

# (i)1 two portfolios tangent to the frontier
xA <- tangency(Rf1) # same portfolio as G in (f)
xB <- tangency(Rf2)
stopifnot(identical(xA, xG))

E_A <- as.numeric(t(xA) %*% Rbar3); sd_A <- sqrt(as.numeric(t(xA) %*% S3 %*% xA))
E_B <- as.numeric(t(xB) %*% Rbar3); sd_B <- sqrt(as.numeric(t(xB) %*% S3 %*% xB))
stopifnot(abs(sum(xB) - 1) < 1e-12)

cat("(i)1 two tangency portfolios\n")
print(round(rbind(A = xA, B = xB), 6))
cat(sprintf("      A: E = %.6f, sd = %.6f   (Rf = %.3f)\n", E_A, sd_A, Rf1))
cat(sprintf("      B: E = %.6f, sd = %.6f   (Rf = %.3f)\n\n", E_B, sd_B, Rf2))

# (i)2 covariance between A and B
sig_AB <- as.numeric(t(xA) %*% S3 %*% xB)

# Merton's constants for the three stocks, used both to check sig_AB against
# the closed form from handout #11 and to check the traced frontier below.
A3 <- as.numeric(t(one3) %*% S3inv %*% Rbar3)
B3 <- as.numeric(t(Rbar3) %*% S3inv %*% Rbar3)
C3 <- as.numeric(t(one3) %*% S3inv %*% one3)
D3 <- B3 * C3 - A3^2
stopifnot(B3 > 0, C3 > 0, D3 > 0)

sig_AB_closed <- (C3 / D3) * (E_A - A3 / C3) * (E_B - A3 / C3) + 1 / C3
stopifnot(abs(sig_AB - sig_AB_closed) < 1e-14)

cat("(i)2 covariance between A and B\n")
cat(sprintf("      xA' Sigma xB                                = %.10f\n", sig_AB))
cat(sprintf("      (C/D)(E_A - A/C)(E_B - A/C) + 1/C          = %.10f\n", sig_AB_closed))
cat(sprintf("      correlation = %.6f\n\n", sig_AB / (sd_A * sd_B)))

# (i)3 trace the frontier out of A and B
#   E_p   = a E_A + (1-a) E_B
#   sd_p^2 = a^2 sd_A^2 + (1-a)^2 sd_B^2 + 2 a (1-a) sig_AB

Elo <- min(E_cloud) - 0.001; Ehi <- max(E_cloud) + 0.001
agrid <- sort(c((Elo - E_B) / (E_A - E_B), (Ehi - E_B) / (E_A - E_B)))
agrid <- seq(agrid[1], agrid[2], length.out = 2000)

fr_E <- agrid * E_A + (1 - agrid) * E_B
fr_sd <- sqrt(agrid^2 * sd_A^2 + (1 - agrid)^2 * sd_B^2 +
             2 * agrid * (1 - agrid) * sig_AB)

# Every mix of A and B must land on the frontier of the three stocks
fr_closed <- sqrt((C3 * fr_E^2 - 2 * A3 * fr_E + B3) / D3)
stopifnot(max(abs(fr_sd - fr_closed)) < 1e-14)

# and the frontier must sit to the left of every point in the cloud
cloud_min_sd <- sqrt((C3 * E_cloud^2 - 2 * A3 * E_cloud + B3) / D3)
stopifnot(all(sd_cloud >= cloud_min_sd - 1e-12))

```

```

cat("(i)3 frontier traced from A and B\n")
cat(sprintf("      max |two-fund mix - closed form| over %d mixes: %.2e\n",
            length(agrid), max(abs(fr_sd - fr_closed))))
cat(sprintf("      min slack of the cloud over the frontier: %.2e\n",
            min(sd_cloud - cloud_min_sd)))

# (i)4 minimum risk portfolio of the three stocks
x3min <- setNames(as.numeric(S3inv %*% one3 / C3), stk3)
E3min <- A3 / C3
sd3min <- sqrt(1 / C3)

stopifnot(abs(sum(x3min) - 1) < 1e-12)
stopifnot(abs(as.numeric(t(x3min) %*% Rbar3) - E3min) < 1e-15)
stopifnot(abs(sqrt(as.numeric(t(x3min) %*% S3 %*% x3min)) - sd3min) < 1e-15)
stopifnot(sd3min <= min(sd_cloud))      # the leftmost attainable point

cat("(i)4 minimum risk portfolio, three stocks\n")
print(round(x3min, 6))
cat(sprintf("      E = A/C = %.6f (%.4f%%)      sd = sqrt(1/C) = %.6f (%.4f%%)\n\n",
            E3min, E3min * 100, sd3min, sd3min * 100))

# ---- plotting helpers ----
# Same palette and helpers as Project 2, so the figures match across projects.

surface <- "#fcfcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_front <- "#2a78d6"      # frontier and the portfolios that sit on it
c_eq    <- "#eb6834"      # tangency portfolio, CAL
c_spx   <- "#1baf7a"      # the traced frontier in (i)3
c_arb   <- "#6f6d67"      # individual stocks

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}
ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
       cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}
pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
}

# ---- (c) and (d): the two stock possibilities curve ----

open_png("fig_two_stock.png")
xr <- c(0.05, max(sd2c) * 1.02); yr <- range(E2c) + c(-0.0006, 0.0006)
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)

eff2 <- E2c >= E2min
lines(sd2c[!eff2], E2c[!eff2], col = c_front, lwd = 1.2, lty = 3)  # inefficient
lines(sd2c[ eff2], E2c[ eff2], col = c_front, lwd = 2.6)          # efficient

```

```

pt(sqrt(vx), Rbar2["XOM"], c_arb, "Exxon-Mobil", 2)
pt(sqrt(vb), Rbar2["BA"], c_arb, "Boeing", 2)
pt(sd2min, E2min, c_front, "minimum risk", 4)

ax(1, pretty(xr), sprintf("%.0f%%", pretty(xr) * 100))
ax(2, pretty(yr), sprintf("%.1f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(c)-(d) Portfolio possibilities curve, Exxon-Mobil and Boeing", 3,
      line = 1.8, adj = 0, col = ink, cex = 1.02, font = 2)
mtext("Solid arm is the efficient frontier; dotted arm is dominated by it.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.75)
dev.off()

# ---- (e), (f), (g): cloud, tangency portfolio, capital allocation line ----

open_png("fig_cal.png", h = 5.2)
# left padding so the labels on the low-risk portfolios are not clipped
xr <- c(-0.04, max(sd_cloud) * 1.03)
xtick <- pretty(c(0, max(sd_cloud)))
yr <- c(min(E_cloud) - 0.0012, max(E_cloud) + 0.0018)
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = xtick, col = grid, lwd = 1)

points(sd_cloud, E_cloud, pch = 16, col = "#c9c8c0", cex = 0.42)

# the capital allocation line:  $E = R_f + \text{slope} * \sigma$ 
abline(a = Rf1, b = slope_G, col = c_eq, lwd = 2)
text(xr[2] * 0.62, Rf1 + slope_G * xr[2] * 0.62, "CAL", pos = 3, offset = 0.4,
     cex = 0.75, font = 2, col = c_eq)

pt(0, Rf1, c_eq, expression(R[f] == 0.001), 4, off = 0.5)
pt(sd_G, E_G, c_eq, "G (tangency)", 4)
pt(sd_g, E_g, c_front, "60% G + 40% Rf", 2)

ax(1, xtick, sprintf("%.0f%%", xtick * 100))
ax(2, pretty(yr), sprintf("%.1f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(e)-(g) Cloud, point of tangency, and the capital allocation line", 3,
      line = 1.8, adj = 0, col = ink, cex = 1.02, font = 2)
mtext(sprintf("%d combinations of Exxon-Mobil, McDonalds and Boeing with weights summing to 1.",
              nrow(X3)), 3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

# ---- (i)3: the frontier traced from A and B, drawn over the cloud -----

open_png("fig_frontier3.png", h = 5.2)
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = xtick, col = grid, lwd = 1)

points(sd_cloud, E_cloud, pch = 16, col = "#333331", cex = 0.42)

```

```

eff3 <- fr_E >= E3min
lines(fr_sd[!eff3], fr_E[!eff3], col = c_spx, lwd = 2.4)
lines(fr_sd[ eff3], fr_E[ eff3], col = c_spx, lwd = 3.2)

pt(sd_A, E_A, c_eq, "A", 2, off = 0.5)
pt(sd_B, E_B, c_eq, "B", 2, off = 0.5)
pt(sd3min, E3min, c_front, "minimum risk", 2)

ax(1, xtick, sprintf("%.0f%%", xtick * 100))
ax(2, pretty(yr), sprintf("%.1f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(i) Efficient frontier traced from portfolios A and B", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.02, font = 2)
mtext("Green curve is combinations of A and B only; it envelopes the cloud from the left.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

cat("wrote fig_two_stock.png, fig_cal.png, fig_frontier3.png\n")

```